

Assignment 5

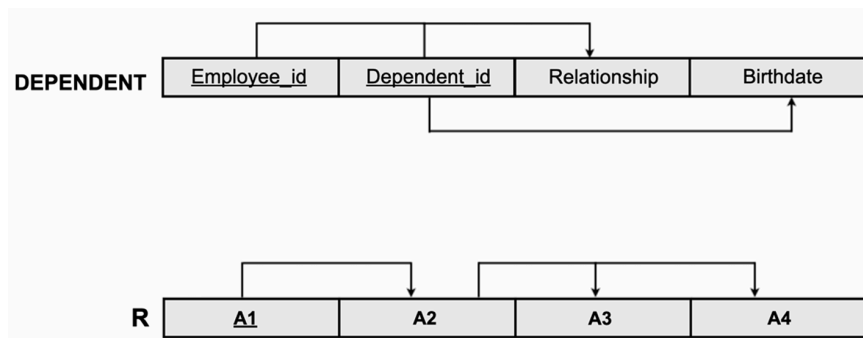
Data Management (INFO125)

Deadline: Friday, 08.11.2024 at 13:00

INSTRUCTIONS: To be completed in groups of 1-3 students. The assignment is mandatory for all students of INFO125 and should be submitted on Mitt UiB. Please note that questions about the assignments or group seminars should be directed to your seminar leader (TA).

Question 1

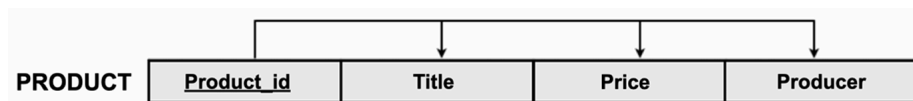
Fill in the blanks by specifying the normal form each of the following relations are in and the reason.



- a) DEPENDENT is in 1 normal form, because of partial dependencies.
- b) Relation R is in 2nd normal form, because of transitive dependencies.

Question 2

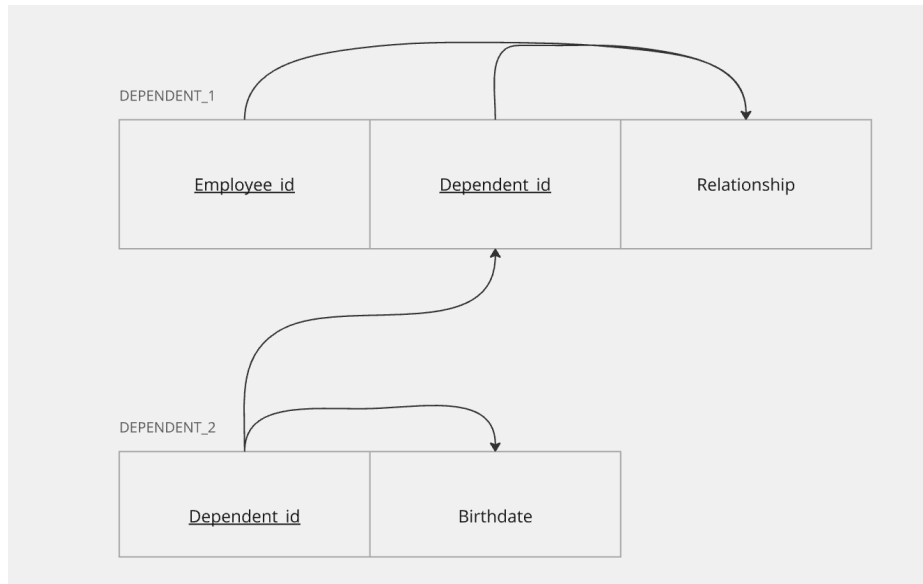
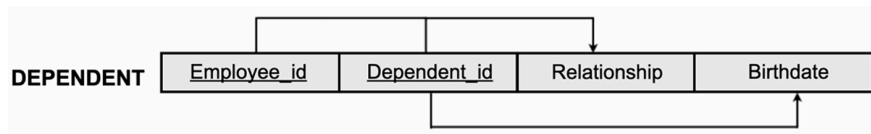
Fill in the blanks by specifying the normal form the following relation is in and the reason.



PRODUCT is in BCNF normal form, because everything relates to product_id.

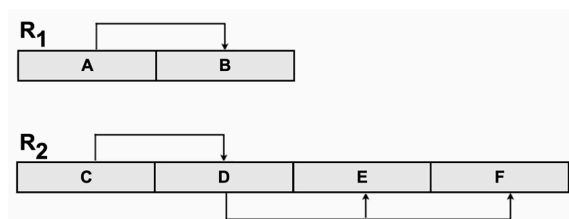
Question 3

Normalize the DEPENDENT relation presented in the following figure.



Question 4

Consider the relations given in the following figure:



Assuming that A is the key for R1 and C is the key for R2, the functional dependencies are:

- f1: {A} → {B}
- f2: {C} → {D}
- f3: {D} → {E, F}

- a) Describe which normal form, each of these relations is in.
 - a. R1 is in BCNF because it meets all the requirements. B only depends on A which is a super key.
- b) For those normal forms that the relations are not in, briefly explain the reason.
 - a. R2 is in 2NF. But it is not in 3NF because $C \rightarrow D$ and $D \rightarrow EF$ are transitive. It is not in BCNF because it is not in 3NF.

Question 5

Suppose that a disk unit has the following parameter:

- block size $B = 1200$ bytes

Consider a STUDENT file with the following fields:

- NUMBER, 10 bytes;
- LASTNAME, 20 bytes;
- FIRSTNAME, 20 bytes;
- BIRTHDATE, 10 bytes;

The STUDENT file has $r=10000$ STUDENT records, fixed-length format, and unspanned blocking. Calculate the following values for the STUDENT file:

- Calculate the record size R
 - 60 bytes
- Calculate the blocking factor bfr
 - $1200/60 = 20$ records per block
- Calculate the number of disk blocks b
 - $10000/20 = 500$ blocks
- Calculate the unused space in each block
 - $20 * 60 = 1200 - (20 * 60)$ bytes = 0, the blocks are full